

SOFT TISSUE INJURIES MODULE

What to remember

Online Module Quick Reference

Each lesson of the online module is distilled to its four or five essential ideas. This document does not replace the manual, the video capsules or in-class practice.

Soft tissue injuries are the most frequent injuries in the backcountry, and often the only ones you will manage from start to finish, with no medical handover. Three principles run through the whole module: control the bleeding, prevent infection, promote healing.

MODULE CONTENTS: SEVEN LESSONS

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Introduction to the chapter

The most frequent injuries in the backcountry, often managed from start to finish by the first aider

Why this chapter: a wound poorly cleaned on day 1 becomes an infection on day 3 and an evacuation on day 5. Here, the first aider does what an emergency physician would do, with what is in their pack.

- 1 Three blocks: understand, manage, equip yourself.** The skin and its response to injury, then assessment through the PAS; specific injuries, blisters and burns; and finally the equipment you will use throughout the chapter.
- 2 Three principles, always in this order.** Control the bleeding, prevent infection, promote healing. Every action in the module ties back to one of the three.
- 3 Shock is never far away.** External or internal hemorrhage produces hypovolemic shock; so does an extensive burn, through plasma loss; an advanced infection, septic shock. Ask yourself in every case: which type of shock?
- 4 Depth is read through the anatomical layers.** Epidermis, dermis, hypodermis, deep fascia. What is damaged beneath the skin (tendon, nerve, vessel, bone, joint) determines the risk and the decision to evacuate, far more than the length of the wound.
- 5 Concepts online, hands-on in class.** Irrigation, packing, dressings, bandages, blister drainage, managing a kit as a pair: all of this is practised in person on moulaged wounds.

LEARNER'S MANUAL

Section 7, Soft Tissue Injuries · Online encyclopedia

FIELD MANUAL

Pages 87 to 101 and 226 to 227

IN CLASS

Wound treatment demonstrated in front of the group, two-rescuer scenario, abrasion and laceration

01

The integumentary system

Understanding the skin to assess what is damaged

The skin is an organ: a barrier against microbes and water loss, a temperature regulator, the organ of touch. Every wound is a breach in these functions, and the body has a plan to close it.

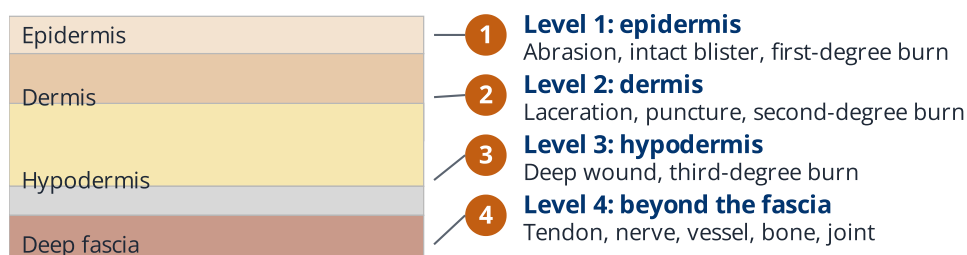


Figure 1 — Four levels, four anatomical boundaries.

- Four levels, four anatomical boundaries.** Epidermis (protective layer, no vessels), dermis (vessels, nerves, glands, follicles), hypodermis (fat, large vessels, cushioning), then the deep fascia covering muscles, tendons, bones and joints. A level 4 injury reaches the deep structures.
- Stopping the bleeding: the body starts without you.** Retraction of severed vessels, vasoconstriction, formation of the platelet plug, then the clot. Your direct pressure gives these mechanisms time to work; it does not replace them.
- Then inflammation, which is normal.** Redness, warmth, swelling and pain in the first days are the body's cleanup response. They become worrying if they increase after 48 hours, spread beyond the wound edges, or come with pus, fever or red streaks.
- Regeneration: moist, oxygenated, nourished, still.** New tissue grows in a moist environment (up to 30% faster than in open air), with a good blood supply, and without being reopened with every movement.
- What hinders healing.** Interrupted blood supply (cold, compression, bandage too tight, smoking), mechanical stress (movement, tension on the edges), infection (abscess, anaerobic infection of deep, closed wounds), and general condition: malnutrition, diabetes, dehydration, exhaustion.

02

Assessment

The PAS applied to a soft tissue injury

A bleeding wound draws the eye; the PAS brings you back to order. Massive hemorrhage is controlled in the primary survey; everything else waits for the secondary survey. And a wound can hide a fracture, a nerve or tendon injury.

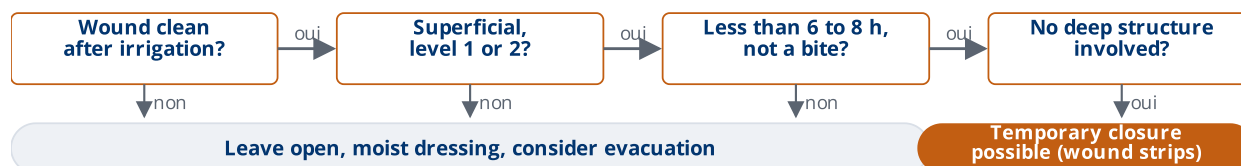


Figure 2 — Close or not in the field: four questions, from the decision aid in the field manual (page 91).

- 1 Describe the wound in four words: force, type, depth, bleeding.** Blunt or penetrating force (firearm: entry, exit, path). Closed or open wound (abrasion, laceration, puncture, avulsion, amputation). Depth level 1 to 4. Capillary, venous or arterial bleeding, and signs of internal hemorrhage.
- 2 Assess what lies beneath the wound.** CSM distal to the injury (circulation, sensation, motion), movement against resistance for tendons, fine sensation for nerves, bone or joint pain on palpation, foreign body. A wound over a joint or directly over a bone is a high-risk wound.
- 3 Control the bleeding.** Firm, continuous direct pressure; if that is not enough, pack the wound with gauze (hemostatic if available) aiming at the bleeding point, removing and repacking if needed to better target the source; tourniquet for uncontrollable limb hemorrhage. Note the time of any tourniquet.
- 4 Prevent infection: irrigate, a lot.** Clean the surrounding skin, then irrigate the inside of the wound with drinking water under pressure (syringe) in large quantity, remove debris with tweezers, explore. Shave if needed, do not scrub the wound bed with antiseptic. A dirty or high-risk wound is not closed in the field.
- 5 Promote healing and decide on evacuation.** Moist wound, dry clean dressing changed as soon as soiled, or occlusive dressing for the long term; wound closure strips only for a superficial, clean, non-bleeding wound. Evacuate if deep structures are involved, heavily contaminated wound, bite, wound of the face, hand or perineum, embedded foreign body, or progressing signs of infection.

03

Specific injuries

Each type of wound has its particularities

The principles do not change; the details do. This lesson covers closed wounds and a few classic field situations that call for a specific action or a specific decision.

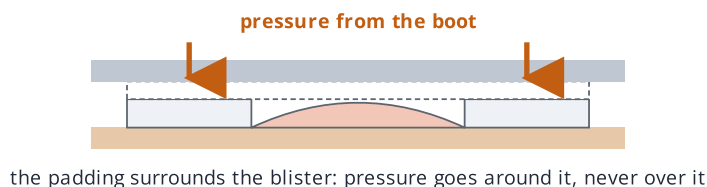
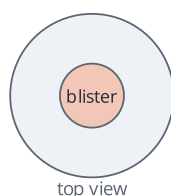
- 1 Contusion: cold, compression, elevation, and a doubt.** A rapidly growing hematoma, disproportionate pain or impaired CSM raise suspicion of a fracture, a vascular injury or compartment syndrome. Reassess.
- 2 Crush injury: the danger is what comes next.** Prolonged compression of a limb releases, upon release, toxins that threaten the heart and kidneys. Assess, immobilize, treat for shock, evacuate. Deep damage can be severe beneath nearly intact skin.
- 3 Ring and swelling: remove it before it is too late.** Any hand or finger injury means removing rings and bracelets immediately, before edema sets in. Thread or string-wrap technique covered in class; cutting the ring as a last resort.
- 4 Subungual hematoma: drain to relieve.** Blood under the nail creates very painful pressure. A small hole in the nail (heated paper clip or needle) releases the pressure, within 24 to 48 hours of the injury.
- 5 Fishhook: push and cut, or remove by traction.** If the point and barb are close to the surface: push through, cut the barb, back the hook out. Otherwise, string technique: pressure on the eye and a sharp pull along the shank. Never near the eye: stabilize and evacuate. Check tetanus status.

04

Blisters

Recognizing the hot spot before it becomes a blister

A friction blister is a soft tissue injury: the upper layers of the epidermis separate under the effect of friction and pressure, and the space fills with fluid. It is the injury that stops the most expeditions, and the most preventable.



Doughnut: thick layer
cut around it, never
placed on the hot spot.

Figure 3 — The padding surrounds the blister: pressure goes around it, never over it.

- 1 **The hot spot is the moment to act.** Redness, warmth, localized burning sensation. Stop, take off the boot, fix the cause (damp sock, lace, wrinkle, gravel, boot too tight) and protect the area before setting off again.
- 2 **Prevention: reduce friction, do not add bulk.** Broken-in footwear, suitable dry socks, anti-friction cream on sensitive areas, thin tape (Adhesive Knit, surgical tape) as prevention. Thick moleskin applied preventively can increase pressure in a snug boot as soon as the foot swells: save padding for blisters that have formed.
- 3 **Intact blister: protect, and drain if it gets in the way.** Small and not tense: leave it, protect with a doughnut dressing. Large, tense or painful when walking: drain cleanly at the base with a disinfected needle, without removing the skin, then dress.
- 4 **Open blister: it is a wound.** Clean, irrigate, keep moist (hydrogel such as Second Skin or hydrocolloid dressing), doughnut dressing around it to spread the pressure, secure with tape that holds when damp. Watch for infection.
- 5 **Long-term management.** Check and redo the dressing every day and whenever it peels off or gets wet. An infected blister (spreading redness, pus, increasing pain, fever) in a hiker can justify an evacuation. Blister kit: needle, antiseptic, hydrogel, thin tape, scissors.

05

Burns

Recognize and classify before treating

A burn is a soft tissue injury caused by heat, cold, a chemical, electricity or radiation. Its severity depends on three things: depth, extent and location. The associated shock is hypovolemic, through plasma loss.

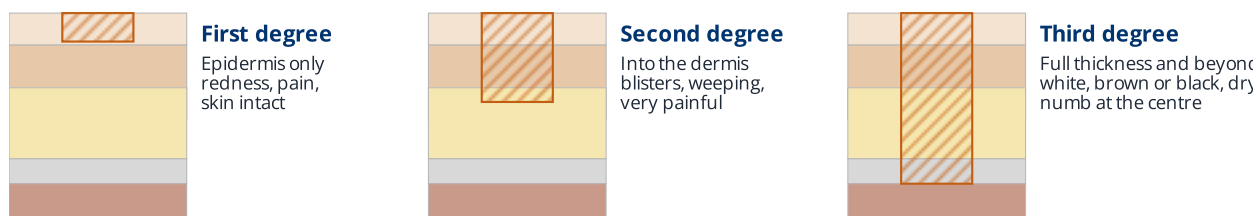


Figure 4 — The shaded depth defines the degree, not the extent.

- 1 **Three degrees, read on the skin.** 1st degree: redness, pain, skin intact (sunburn). 2nd degree: blisters, weeping, very painful, reaches the dermis. 3rd degree: white, brown or black, dry, numb at the centre, through the full thickness of the skin. Classification is based on clinical criteria, not on a percentage of surface area.
- 2 **First aid is the same for all of them.** Move away from the source, universal precautions, ABC. Cool with plenty of cool (not ice-cold) water, remove non-adherent clothing, rings and jewellery before swelling. Never remove anything that is stuck. Snow and ice with great caution.
- 3 **According to degree.** 1st degree: moisturizer such as aloe on intact skin. 2nd and 3rd degree: irrigate gently, do not break blisters, no cream on an open blister, occlusive burn dressing (hydrogel) or dry sterile gauze. Antibiotic ointment only if evacuation to care exceeds 24 hours.
- 4 **Hydration, warmth, pain.** An extensive burn loses plasma and heat: have the patient drink as much as tolerated, treat for shock and prevent hypothermia. Applied cold relieves pain but cools the patient: cool the burn, warm the person.
- 5 **When to evacuate.** 2nd degree over more than 15% of body surface area (10% in children or older adults), extensive 3rd degree, facial burn or signs of inhalation (cough, soot, hoarse voice, singed nasal hairs), deep burns of the hands, feet, genitals, eyes or joints, chemical or electrical burns (invisible internal injuries, arrhythmias).

06

First aid kits

The equipment you will use throughout the chapter

A kit is not a list, it is a plan. It is built according to duration, number of participants, level of training, environment and anticipated risks. And it is only useful if you know it before you need it.

- 1 **Five families of supplies for soft tissue.** Sterile dressings (gauze pads, non-adherent gauze), non-sterile dressings (padding), bandages (roller gauze, elastic, triangular), adhesives (cloth tape, surgical tape, tincture of benzoin), specialty dressings (wound closure strips, transparent occlusive, hydrogel, hydrocolloids).
- 2 **Cleaning tools.** Irrigation syringe, fine tweezers, scissors, plenty of gloves, alcohol-free antiseptic solution for the surrounding skin, drinking water for the inside of the wound. A water bottle with a squirt cap makes a good improvised irrigator.
- 3 **Choose your kit according to your activity.** Canoeing, climbing, snowmobiling, worksite, hiking: the mechanisms of injury differ, and so does the content. Think of blisters, stove burns, hands. Pack in waterproof, labelled modules.
- 4 **Get familiar with it before the in-person module.** Open every dressing at least once, know how it is applied and where it is stored. In class, you will manage the kit as a pair: one treats, the other supplies and keeps the equipment clean.
- 5 **Managing the kit during and after.** Do not contaminate the kit with soiled gloves, isolate waste, clean the tools, take inventory after every use, note what is missing. A kit that has been opened and not restocked is false security.